Specification

Introduction and Project Context

Intrinsically disordered proteins (IDPs) lack stable three-dimensional structure under physiological conditions but play crucial roles in numerous biological processes, including gene regulation, cell signaling, and molecular recognition [1]. Despite being heavily represented in the proteome, accounting for approximately 30% of eukaryotic proteins [13], the prediction of binding sites in these disordered regions has only recently become a focus of systematic research.

The development of computational methods for binding site prediction in structured proteins has a longer history, with well-established tools such as fpocket [14], developed in the late 2000s, leveraging geometric and structural features to identify binding pockets in stable protein structures. However, predicting binding sites in IDPs presents unique challenges due to their inherent flexibility and lack of stable structure, requiring fundamentally different computational approaches.

The Critical Assessment of Intrinsic Disorder (CAID) challenge has emerged as the primary benchmark for evaluating IDP-related predictions. The challenge now includes a dedicated track for binding site prediction, focusing on DNA/RNA binding, protein binding, and ion binding sites in disordered regions [2, 3]. Three rounds of the CAID binding site prediction challenge have been completed, establishing standardized benchmarks and evaluation protocols for the field.

This research project is dedicated to advancing computational approaches for IDP function prediction, with a specific focus on improving binding site prediction through novel hybrid training strategies that will be evaluated against the CAID3 benchmark.

Protein Language Models in Structural Biology

Protein Language Models (pLMs) are deep learning models trained on vast collections of protein sequences to learn the fundamental patterns and relationships encoded in protein sequences. These models, inspired by natural language processing techniques, have revolutionized various protein-related tasks, including structure prediction, function annotation, and binding site identification [15].

Several pLMs have been developed over recent years, ranging from earlier models like ProtBERT to more recent and sophisticated architectures. The current state-of-the-art includes

ESM2 (Evolutionary Scale Modeling 2) developed by Meta AI [16] and the ProtT5 family from RostLab [17]. Both ESM2 and ProtT5 have demonstrated comparable performance across various protein prediction tasks [18], with ESM2 being particularly widely adopted due to its robust performance and accessibility.

For this project, we will utilize ESM2 as our primary feature extraction tool. ESM2 generates contextual embeddings for each residue in a protein sequence, capturing both local and global sequence patterns that are crucial for binding site prediction. Unlike structure-based approaches that rely on 3D coordinates, pLM-based methods can operate directly on sequence information, making them particularly suitable for IDPs where stable structures are absent.

Project Goal and Hybrid Approach

The project's main objective is to improve IDP binding site prediction by developing a hybrid training approach that combines binding site data from both intrinsically disordered proteins and well-structured proteins.

The standard approach for participation in the CAID challenge involves training prediction models exclusively on IDP training data from databases like DisProt [5]. However, our research proposes an alternative strategy: we will train our models not only on IDP binding sites but also incorporate binding site data from proteins with well-defined, stable structures (where the binding sites themselves are structured). This data will be sourced from structural databases such as AHoJ-DB [4].

The rationale behind this hybrid approach is to investigate whether the additional information from structured binding sites can improve the model's ability to recognize binding patterns in disordered regions. By testing this hypothesis systematically, we aim to determine whether supplementing limited IDP training data with abundant structured protein data leads to performance improvements.

To achieve this goal, we will design and test multiple dataset configurations that integrate structured protein binding sites into IDP training sets at varying proportions, and evaluate whether these combined approaches improve prediction performance. Our main test set will be the official CAID3 test set, allowing direct comparison with methods that participated in the most recent CAID challenge round.

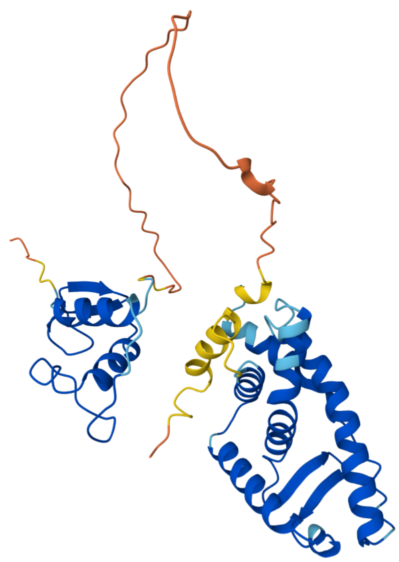
This work leverages the Computational Structural Biology Group's existing expertise and tools, including P2Rank [11] for binding site prediction and insights from the CryptoBench dataset [12] regarding dynamic binding mechanisms.

Structured vs. Disordered Regions

To illustrate the challenge this project addresses, consider the comparison between structured and unstructured protein regions:

AlphaFold prediction of the human Estrogen receptor (DisProt: DP00074). The blue parts indicate the **structured region**, while the dynamic red/orange parts indicate the **disordered region** (a prediction that has been experimentally confirmed).

This visual demonstrates the complexity of IDPs and highlights why specialized methods are necessary for accurately predicting binding sites in these flexible regions.



Why Structure-Based Methods Don't Work for IDPs

The prediction of binding sites in structured proteins can leverage rich structural features. Methods for structured proteins analyze the local 3D neighborhood around each residue, examining geometric properties such as pocket depth, surface curvature, and residue packing. Additionally, these methods can employ structure-based deep learning architectures, such as 3D convolutional networks [19], which directly process the spatial arrangement of atoms and residues.

However, these structure-based approaches are fundamentally incompatible with IDP binding site prediction because disordered regions lack stable three-dimensional structures; only the dynamic parts visible in disorder predictions exist. Without fixed atomic coordinates and stable structural contexts, geometric and spatial features cannot be reliably computed or utilized.

Therefore, for IDP binding site prediction, we must rely exclusively on sequence-based approaches. Protein language models like ESM2 provide a powerful alternative by extracting rich feature representations directly from amino acid sequences, capturing evolutionary patterns and functional signals without requiring structural information. This makes pLMs ideally suited for the challenge of predicting binding sites in intrinsically disordered regions.

Draft of Technical Solution

Tools and Technologies

Programming Languages: Python
Frameworks and Libraries: PyTorch, ESM2 (Meta AI), Hugging Face transformers, pandas, numpy, scikit-learn, BioPython, biotite
Research Tools: DisProt database, UniProt database, AHoJ DB, CAID challenge benchmarks, mmseqs2 tool
Hardware Requirements: GPU access for model training (University Cluster - GPU Lab)

Implementation Strategy

Implementation follows a modular approach with separate components for data processing, feature extraction using transfer learning with ESM2, model training, and evaluation. The system will train specialized models for individual binding site types and unified models for comparison. Code will be organized in a reproducible research framework with version control and documentation. Sequence similarity-based dataset splitting will prevent information leakage between training, validation, and test sets.

Evaluation Metrics

Model performance will be assessed using multiple complementary metrics to provide a comprehensive evaluation:

- **AUC (Area Under the ROC Curve)**: Measures the model's ability to discriminate between positive and negative classes across all classification thresholds
- **AUPRC (Area Under the Precision-Recall Curve)**: Particularly important for imbalanced datasets, focuses on performance for the minority (positive) class
- **MCC (Matthews Correlation Coefficient)**: Provides a balanced measure that accounts for true and false positives and negatives, suitable for imbalanced datasets
- **F1 Score**: Harmonic mean of precision and recall, balancing the trade-off between these metrics
- **Recall**: Measures the proportion of actual binding sites correctly identified

- **Accuracy**: Overall proportion of correct predictions, though less informative for imbalanced datasets

These metrics will be computed on both validation sets during development and on the CAID3 test set for final evaluation and comparison with published methods.

Data Overview and Scope

This project involved the analysis of a substantial dataset to develop robust prediction models. Key statistics of the primary dataset include:

Dataset Type	Source	Proteins	Total Residues	Binding Sites	Positive %	Annotations
Static Binding Sites						
Ion Binding	AHoJ-DB	173,067	60,799,629	913,380	1.5%	-
IDP Binding Sites						
Protein Binding	DisProt	774	482,596	115,357	23.9%	6,820
DNA Binding	DisProt	71	44,499	11,827	26.6%	726
RNA Binding	DisProt	68	54,326	11,666	21.5%	844
Ion Binding	DisProt	79	38,822	12,156	31.3%	1,128
Summary						
Total DisProt	-	926 unique	620,243	151,006	24.3%	9,518
Combined Dataset	Both	~174,000	~61.4M	~1.06M	1.7%	-

Key Dataset characteristic

Static Binding Sites (AHoJ-DB):

- **Proteins Analyzed:** 173,067 proteins
- **Total Residues:** 60.8 million residues
- **Confirmed Ion Binding Sites:** 913,380
- **Class Imbalance:** Extreme imbalance with 98.5% negative and 1.5% positive examples
- **Ion Types:** 10 different ion types including Zinc (Zn), Magnesium (Mg), Calcium (Ca), Manganese (Mn), Iron (Fe), and Copper (Cu)

IDP Binding Sites (DisProt):

- **Total unique IDP proteins:** 926
- **Total DisProt annotations:** 9,518
- **Covers four binding categories:** protein, DNA, RNA, and ion binding

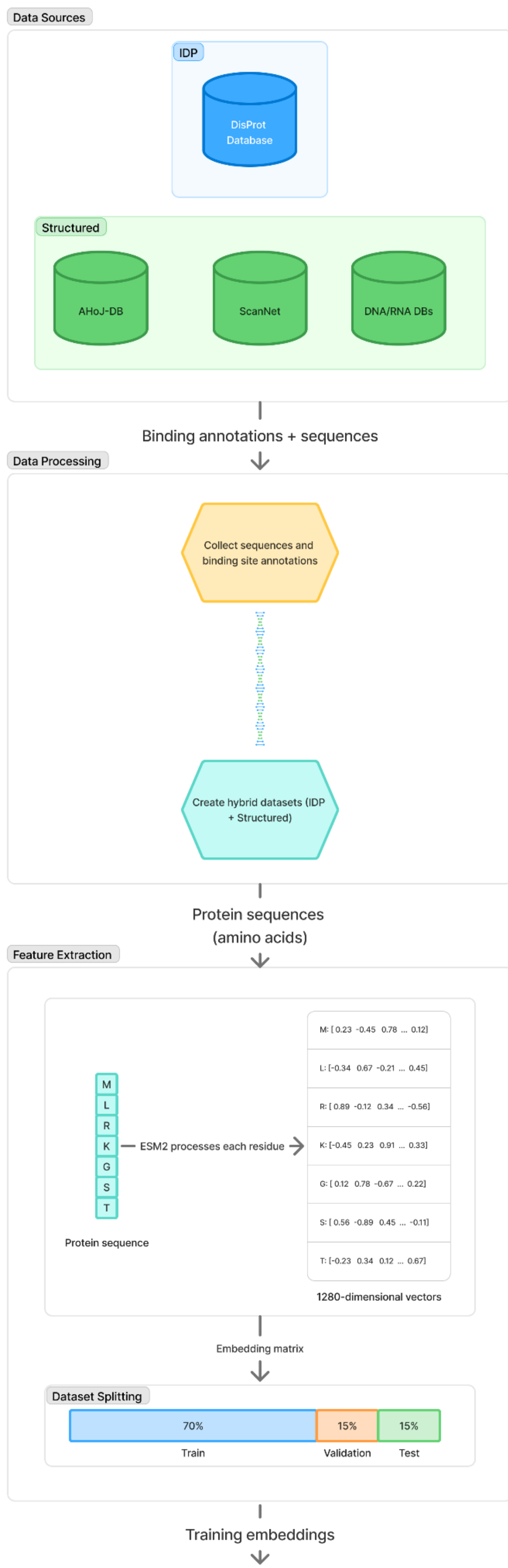
Data Integration Strategy:

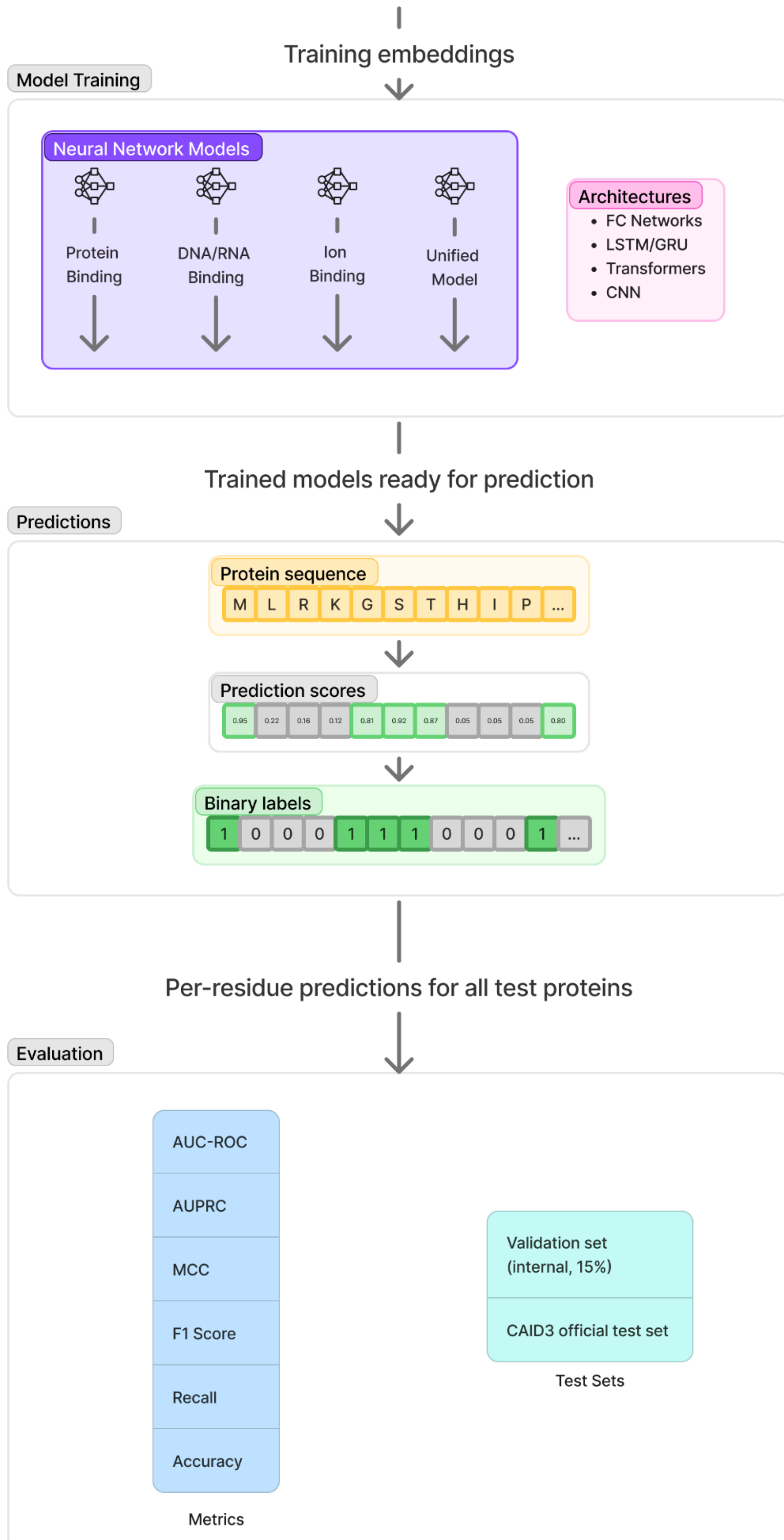
- **Train/Val/Test split:** 70%/15%/15% at cluster level
- **Sequence similarity threshold:** <10% between splits (using mmseqs2 clustering)
- **Main test set:** CAID3 official test set for direct comparison with challenge participants

Detailed Specification of Individual Tasks

The Diagram (Figure 1) provides a visual overview of the complete computational pipeline. The subsequent sections detail each component of this workflow.

Figure 1: Computational workflow for hybrid IDP binding site prediction. The pipeline integrates binding site data from both intrinsically disordered proteins (blue) and well-structured proteins (green). Protein sequences are processed through ESM2 to generate per-residue embedding vectors,numerical representations that capture sequence patterns. These embeddings are then split into training, validation, and test sets based on sequence clustering to prevent data leakage. Neural network models learn to map embeddings to binding site probabilities, which are thresholded to produce binary predictions for each residue. Models are evaluated using multiple metrics on the CAID3 benchmark, enabling direct comparison with state-of-the-art methods.


Data Sources
IDP
DisProt Database
Structured
AHoJ-DB
ScanNet
DNA/RNA DBs
Binding annotations + sequences
Data Processing
Collect sequences and binding site annotations
Create hybrid datasets (IDP + Structured)
Protein sequences (amino acids)
Feature Extraction
M
L
R
K
G
S
T
Protein sequence
ESM2 processes each residue
M: [0.23 -0.45 0.78 ... 0.12]
L: [-0.34 0.67 -0.21 ... 0.45]
R: [0.89 -0.12 0.34 ... -0.56]
K: [-0.45 0.23 0.91 ... 0.33]
G: [0.12 0.78 -0.67 ... 0.22]
S: [0.56 -0.89 0.45 ... -0.11]
T: [-0.23 0.34 0.12 ... 0.67]
1280-dimensional vectors
Embedding matrix
Dataset Splitting
70%
15%
15%
Train
Validation
Test
Training embeddings


Training embeddings
Model Training
Neural Network Models
Protein Binding
DNA/RNA Binding
Ion Binding
Unified Model
Architectures
• FC Networks
• LSTM/GRU
• Transformers
• CNN
Trained models ready for prediction
Predictions
Protein sequence
M L R K G S T H I P ...
Prediction scores
0.95 0.22 0.16 0.12 0.81 0.92 0.87 0.05 0.05 0.05 0.80
Binary labels
1 0 0 0 1 1 1 0 0 0 1 ...
Per-residue predictions for all test proteins
Evaluation
AUC-ROC
AUPRC
MCC
F1 Score
Recall
Accuracy
Metrics
Validation set (internal, 15%)
CAID3 official test set
Test Sets

Task 1: Data Collection and Preprocessing
Description: Collect binding site data from databases of static binding sites and databases of IDP binding sites:

- **IDP binding sites:** DisProt database for intrinsically disordered proteins, covering protein, DNA, RNA, and ion binding annotations in disordered regions.
- **Static binding sites:** Extract from structural databases including:
    - AHoJ-DB archive for ion binding sites in structured proteins [4]
    - ScanNet and Nature publications for protein binding sites [7, 8]
    - Curated datasets for DNA/RNA binding in structured regions [9]
- **Integration:** Design train/test datasets for each binding category with multiple configurations:
    - IDP-only datasets (standard CAID approach)
    - Structured-only datasets (baseline comparison)
    - Hybrid datasets with varying proportions of IDP and structured data
- **Quality control:** Ensure no information leakage by maintaining maximal sequence similarity between training, validation, and test sets below 10% using mmseqs2 clustering tool [10]
- **Test set preparation:** Obtain and format the CAID3 official test set for final evaluation

Deliverables: Complete dataset collection with IDP-only, structured-only, and multiple hybrid configurations for all four binding categories (protein, DNA, RNA, ion), plus properly formatted CAID3 test set
Duration: 2 months

Task 2: Feature Extraction
Description: Extract ESM2 protein language model embeddings for all collected sequences from both IDP and structured protein datasets. This will be performed on the University GPU Lab cluster due to computational requirements:

- Process all sequences through the ESM2 model to generate per-residue embeddings
- Extract embeddings at the appropriate layer (typically the final or penultimate layer)
- Store embeddings in efficient format for downstream model training
- Generate embeddings for training, validation, and test sets

Deliverables: Complete ESM2 embedding datasets for all sequences, organized by dataset configuration and binding category, ready for machine learning model training
Duration: 1 month

Task 3: Neural Network Model Development
Description: Design and implement neural network architectures that operate on ESM2 embeddings using a transfer learning approach. The models will take pre-computed ESM2 features as input and learn to predict binding sites:

- Develop both specialized models trained on single binding categories (protein-protein, DNA/RNA, or ion binding) and unified models that do not differentiate between binding types
- Implement various architectural patterns:
    - Fully connected networks for baseline performance
    - Recurrent networks (LSTM/GRU) to capture sequential dependencies
    - Transformer-based architectures for attention-based sequence modeling
    - 1D convolutional networks for local pattern detection
- Design training strategies to handle extreme class imbalance (particularly for structured protein datasets)
- Implement model checkpointing, early stopping, and other training optimizations

Deliverables: Implemented neural network models with complete training pipelines, hyperparameter configuration files, and training scripts
Duration: 1 month

Task 4: Model Training and Evaluation
Description: Train models on different dataset configurations and conduct comprehensive comparative analysis:

- Train three categories of models:
    - IDP-only models (standard CAID approach)
    - Structured-only models (baseline comparison)
    - Hybrid models with varying proportions of IDP and structured data
- Perform comparison across all configurations to identify optimal data integration strategies
- Conduct cross-validation during development to tune hyperparameters and assess generalization
- Evaluate all models on the CAID3 official test set for final performance assessment and direct comparison with published CAID3 participants
- Analyze performance differences across binding site types (protein, DNA, RNA, ion)
- Investigate which structural data integration strategies (if any) improve IDP binding site prediction

Deliverables: Trained models with comprehensive performance analysis, detailed comparison of dataset configurations, evaluation results on CAID3 test set with comparison to published methods, and analysis of which approaches work best for IDP binding site prediction
Duration: 3 months

Task 5: Research Documentation
Description: Document methodology, results, and conclusions. Prepare a research report and presentation.
Deliverables: Complete research project report and defense presentation
Duration: 2 months

Time Schedule and Milestones

Month	Milestone	Tasks	Deliverables
Month 1-2	Complete Dataset Ready	Task 1	Hybrid datasets with dynamic+static binding sites
Month 3	Feature Extraction Complete	Task 2	ESM2 features for all datasets
Month 4	Model Development Complete	Task 3	Neural network implementations
Month 5-6-7	Training & Evaluation Complete	Task 4	Trained models and performance analysis
Month 8	Research Documentation	Task 5	Research project report
Month 9	Final Submission	Task 5	Final report and defense preparation

Expected Use of Project Output for Research Activities

Research Contribution

This project contributes to computational biology research by developing novel hybrid training methods for IDP binding site prediction. The approach offers insights into combining structured and disordered protein data, potentially applicable to other protein prediction tasks.

Publication Potential

The main goal is to participate in the CAID binding site prediction challenge. Results may be suitable for extension into conference papers or journal publications, with potential venues including bioinformatics conferences and journals.

Tool/Software Contribution

The developed prediction tools will be made available to the research community through GitHub, potentially contributing to future CAID challenges and IDP research efforts.

Expected Impact

The project advances IDP function prediction capabilities, with potential applications in drug discovery and disease research where IDPs play crucial roles. Results will be disseminated through the CAID challenge participation and open-source software release.

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